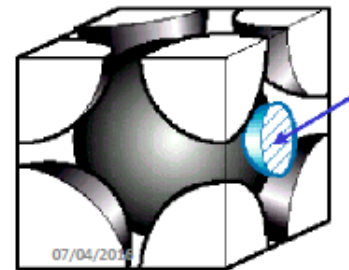
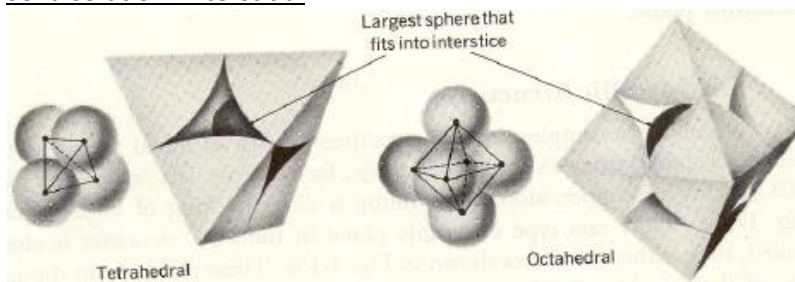


Ni-Cu is the best combination as they have mutual dissolution: Ni can dissolve in Cu and Cu can dissolve in Ni. The valence factor fails for some systems. However, it doesn't mean that the formation of alloys is not possible.

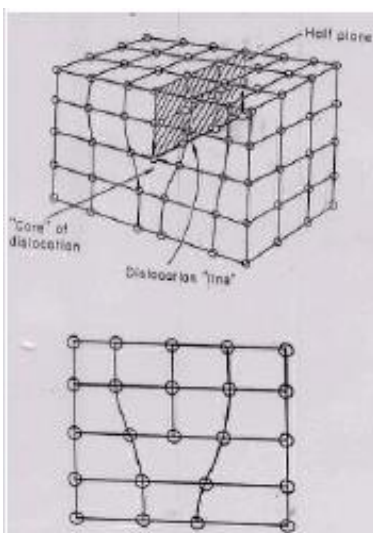
#### Solid solution: Interstitial



We can consider the diffusion of C into Fe as an example. As the size of C atom is small, it will fit with the interstitial cavity. It generates a very severe contraction in the lattice of the Fe BCC in the close neighborhood of the interstitial site as well as in regions further from it.

The max %C in Fe is 2% at room temperature.

#### Linear defects: dislocations, jogs, kinks, disclinations, etc.



Dislocations are linear defects. There exist two basic d. types: **edge** and **screw** d. Interatomic bonds are remarkably **distorted** nearby the dislocation line, the so called '**d. core**'.

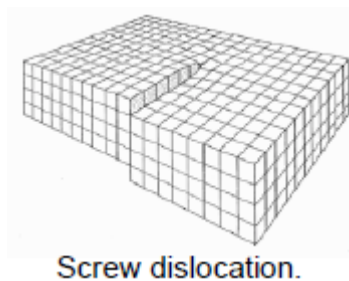
They induce **intensive and extensive elastic strains** in the lattice, several lattice spacings away.

The basic concept of dislocation was introduced for the first time in 1909 by **Volterra**, an Italian scientist in his treatise on *Elastic theory in defective media*. Later dislocations were utilized in 1939 independently by Taylor, Orowan and Polyani to explain the mechanical properties in real metals and alloys, especially their plastic behavior at room temperature.

Dislocations represent the physical means to describe the plastic behavior of ceramics at high temperature; they have a rotational movement around x, y and z axis.

To characterize dislocations, the **Burgers' vector** is used.

#### Edge, screw and mixed dislocations



The Burgers' vector is obtained by tracing a circuit around the dislocation core. When there is a dislocation, the circuit doesn't close. In this open space of the circuit we have the Burgers' vector.

Depending on the direction of the Burger's vector with respect to the d. line (*d.l.*) either the edge or screw dislocation can be discerned.

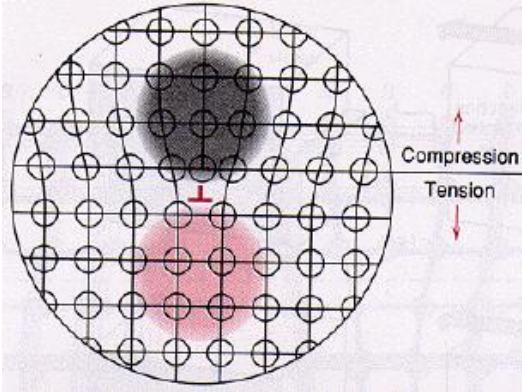
For the **edge d.**:  $b \perp (d.l.)$

For the **screw d.**:  $b \parallel (d.l.)$

where *d.l.* represents the direction along which the material is deformed.

The edge and screw dislocations constitute the two simplest lattice building blocks in the metals to promote plastic deformation. In reality, the morphology of defects originated during plastic deformation is much more complex and usually **mixed d.**, i.e. which have mixed edge and screw character, can be observed. In addition, the amount of mixed character may change continuously during plastic deformation, due to their motion and interference with other defects.

## Stress field around dislocations



Any dislocation produces a stress field.

As an **edge d.** moves through the lattice, it creates a stress field around it, consisting of both a **compressive** and a **tensile stress field** in addition to **shear stress field** along the d. core.

Specifically, the atoms above the edge d. are compressed and those below are moved further and abnormally from the d.l. establishing a tensile stress field.

Around the d.l. a shear region is produced.

A **screw d.**, on the other hand, produces a **shear deformation**.

Interaction of dislocations → formation of vacancies and

interstitials, depending on the side along which the dislocation is moving.

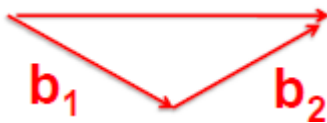
⊥ ⊥      ⊥ ⊥ The interaction of dislocations with the same side will produce a train of vacancies.

v      i      The interaction of dislocation with opposite sides will produce a train of interstitials.

## Dissociation of a perfect dislocation

Screw and edge dislocations are ideal dislocations and they are very rare. In order to cross the atom, atom has to be sheared, which is very expensive. It is easier to surround the atom rather than sharing it by changing the directions.

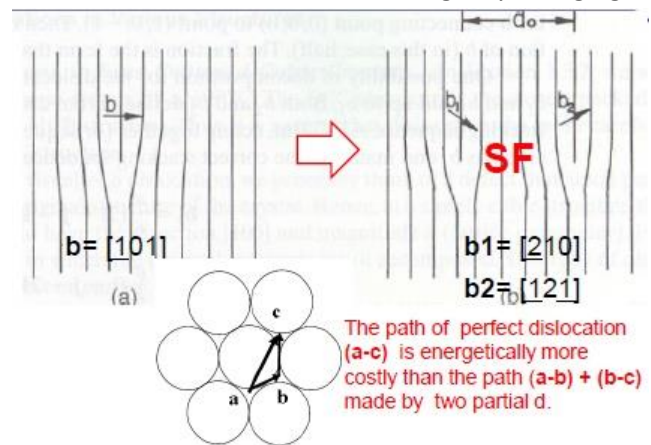
Perfect d. spontaneously decomposes into partial dislocations whose core may propagate over much larger regions consuming lower energy and interacting with more lattice defects than those met by the original dislocation.



Two partial dislocations have  $b_1$  and  $b_2$  Burgers' moduli respectively and their summation gives us the original Burgers' vector.

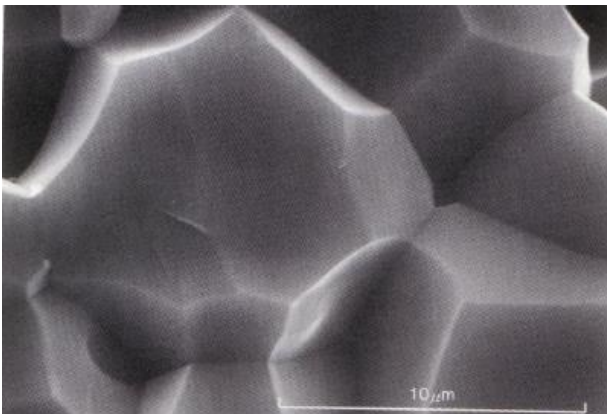
Any ideal dislocations will move by relying on two partial dislocations and whenever partial dislocation produced also as stacking fault will be formed. Whenever we have a stacking fault, the FCC packing will not be a,b,c,a,b,c... but in some parts we will have a,b,a,b : Stacking fault : one plane will be missing.

Stacking faults are mostly found in FCC while BCC is usually not affected.



The path of perfect dislocation (a-c) is energetically more costly than the path (a-b) + (b-c) made by two partial d.

**2D defects:** Grain boundaries, twin boundaries, stacking fault, CSL boundaries, interfaces (coherence, incoherence, semicoherence), interphases, free surfaces, etc.



Grain Boundaries by Optical Microscope

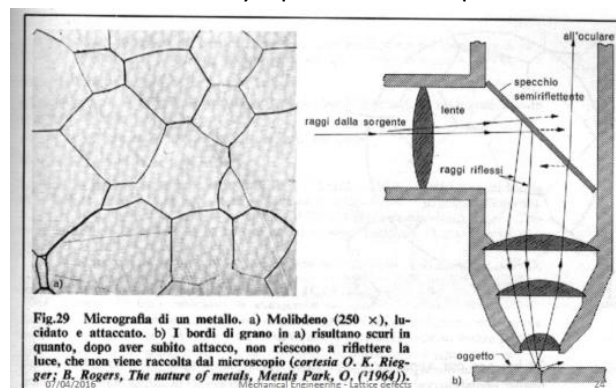
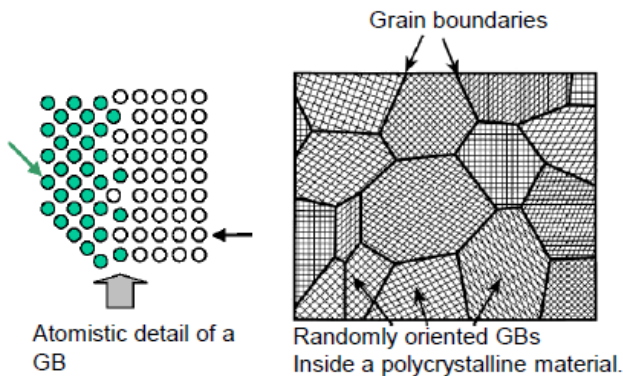


Fig.29 Micrografia di un metallo. a) Molibdeno (250 ×), lucidato e attaccato. b) I bordi di grano in a) risultano scuri in quanto, dopo aver subito attacco, non riescono a riflettere la luce, che non viene raccolta dal microscopio (cortesia O. K. Rieger; B. Rogers, The nature of metals, Metals Park, O. (1964)).

By using a corrosive acid, grain boundaries become black (corroded) because of the reactivity since they have high energy → I can isolate the grains and see their size. Grains are in a more stable conditions, the boundaries are more disordered (higher energy and higher reactivity).

#### Polycrystalline materials: GBs



GBs represent boundaries regions in which different individual grain orientations are accommodated. It is a high energy region, due to the disordered arrangement of atoms.

#### Polycrystalline materials: the role of free surfaces and interfaces

Superficial atoms have incomplete bonds and greater energy than internal ones → the overall surface energy tends to be minimized by minimizing the superficial area (e.g. water drop falling in air).

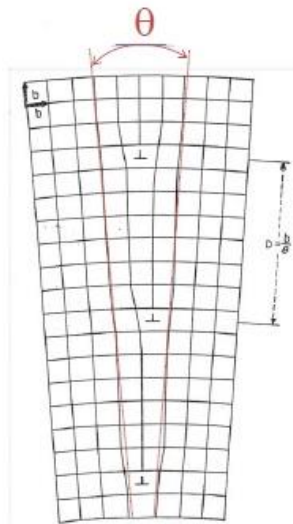
The solid surfaces rearrange to fulfill superficial atomic bond requirements.

Surface and interfaces strongly react with impurities and easily capture them (segregation).

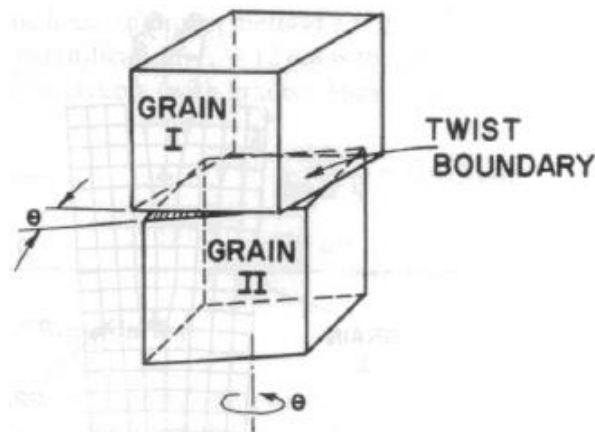
As there exists an associated energy with surfaces (e.g. GBs) and bulk grains, the latter tend to grow at the expense of smaller ones, in the attempt to minimize their interfacial energy.

This is a diffusion process, thus can be significantly accelerated at higher temperatures (grain growth).

#### Tilt and Twist GBs: low angle GBs ( $<15^\circ$ )



**Tilt boundary** (adjacent co-planar grains, described by a sequence of dislocations)



**Twist boundary** (adjacent superposed grains)

#### Grain boundaries: main features

They are 2D defects, typical in polycrystalline materials.

They are produced by the relative rotation of two adjacent grains.

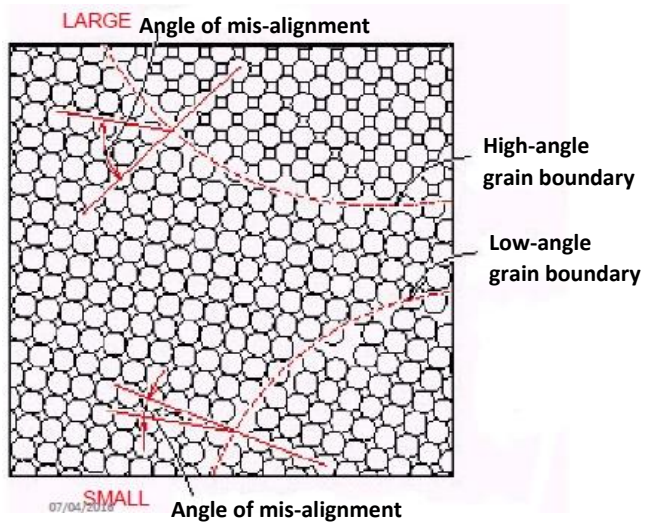
The rotational angle can be:

- Small (if  $\leq 15^\circ$ ) **small-angle GBs**
- Large (if  $\geq 15^\circ$ ) **large-angle GBs**

When the angle is small, the GB can be viewed as constituted by a sequential array of edge dislocations.



In large-angle GBs, other than edge dislocations there might simultaneously exist other defects at the GB, therefore it is not correct anymore its description by only edge dislocations.



Depending on the value of the mis-alignment angle between atomic planes of two adjacent grains, a GB can be divided into a small- and large-angle GB

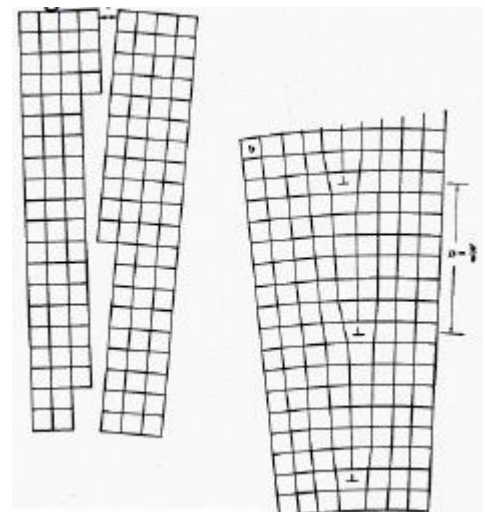
The angle between two adjacent grains is named **misorientation angle ( $\theta$ )**.

This angle arises when two crystalline grains are relatively rotated of an amount equal to  $\theta$  angle with respect to an axis being perpendicular to both crystal planes.

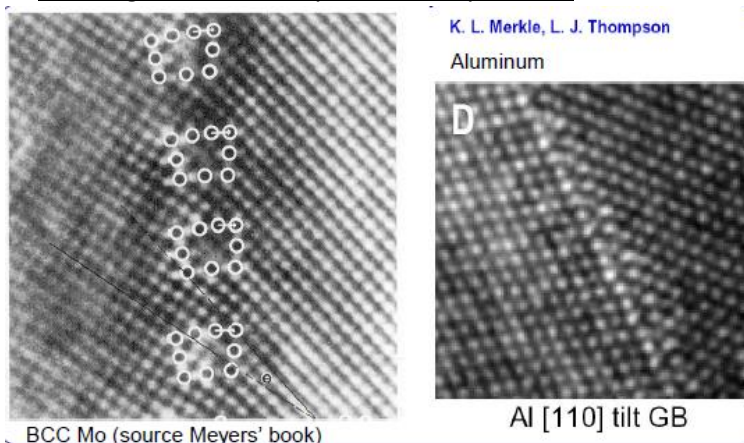
The junction between the two grains is absorbed by the common GB through the equivalent interposition of an array edge dislocations separated on an amount of  $b/\theta$  apart.

The energy associated to a low-angle GB is much smaller than that attributed to a large-angle GB.

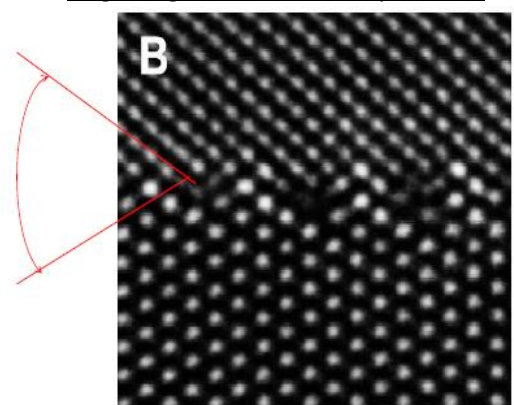
GBs constituted by edge dislocations are named **Tilt boundaries**.



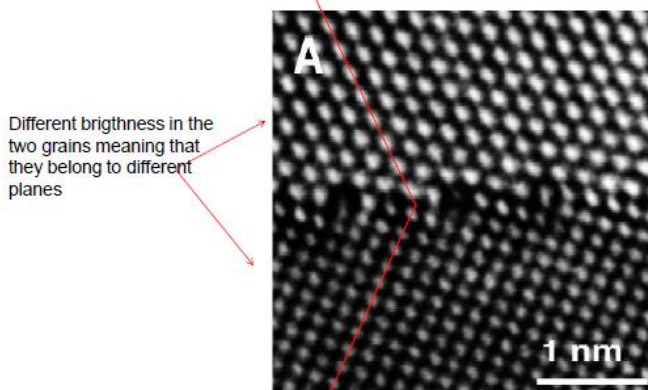
#### Low-angle Tilt boundary observed by HRTEM



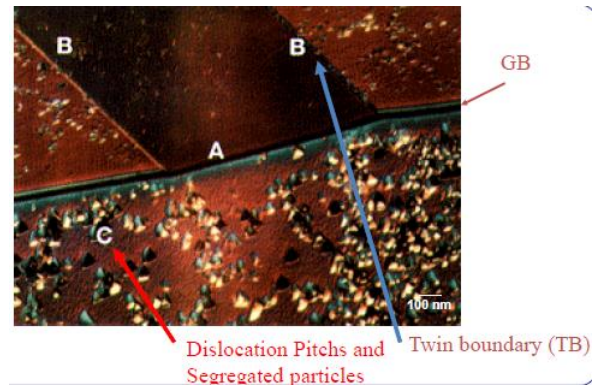
#### High-angle Tilt GB in Au by HRTEM



### High angle Twist GB in Au by HRTEM



### Nanososcopic Structure taken by TEM

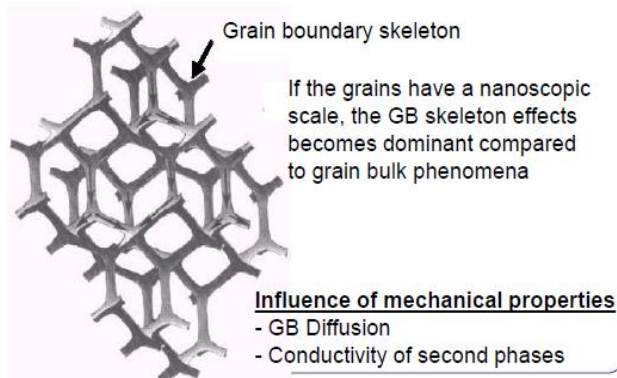


The dislocation pitch can be observed also by etching at optical microscope.

### GB Skeleton of a polycrystal

Diffusion is an activating process → for a fast diffusion, we need energy for a successful jump.

The channels of the GB skeleton act as fast diffusion paths.



The deviation of atom positions from the perfect lattice leads to a higher energy state in GBs.

In case of small grains → higher energy GB → faster diffusion at GB.

Creep resistance is lower when grains are small and larger when grains are large. When we have creeps, we can't reduce dislocations because high temperature will make plastic flow easy.

### Coincidence Site Lattice (CSL) GBs

With increasing misorientation angle between two crystals the surface energy increases. Nevertheless, there exist a special class of large-angle GBs which are characterized by a significantly low surface energy. The relative rotation of the two grains lead to make a fraction of the atoms of both grains coinciding along the common boundary. Such a special GB is named, **Coincidence Site Lattice (CSL)** GB. The fraction of common atoms between the two grains is measured by the  $\Sigma$  parameter, which is reciprocal to fraction of common sites. For instance, a twin boundary (TB) can be viewed as a CSL GB with  $\Sigma=3$ , meaning that the two adjacent grains have a common site every 3 atoms being counted along the common boundary. The CSL GBs having  $\ll \Sigma$  may remarkably affect the mechanical properties of polycrystalline materials. The CSL GBs with  $\Sigma < 29$  offer great advantages compared to conventional large-angle GBs (with  $\gg \Sigma$ ).

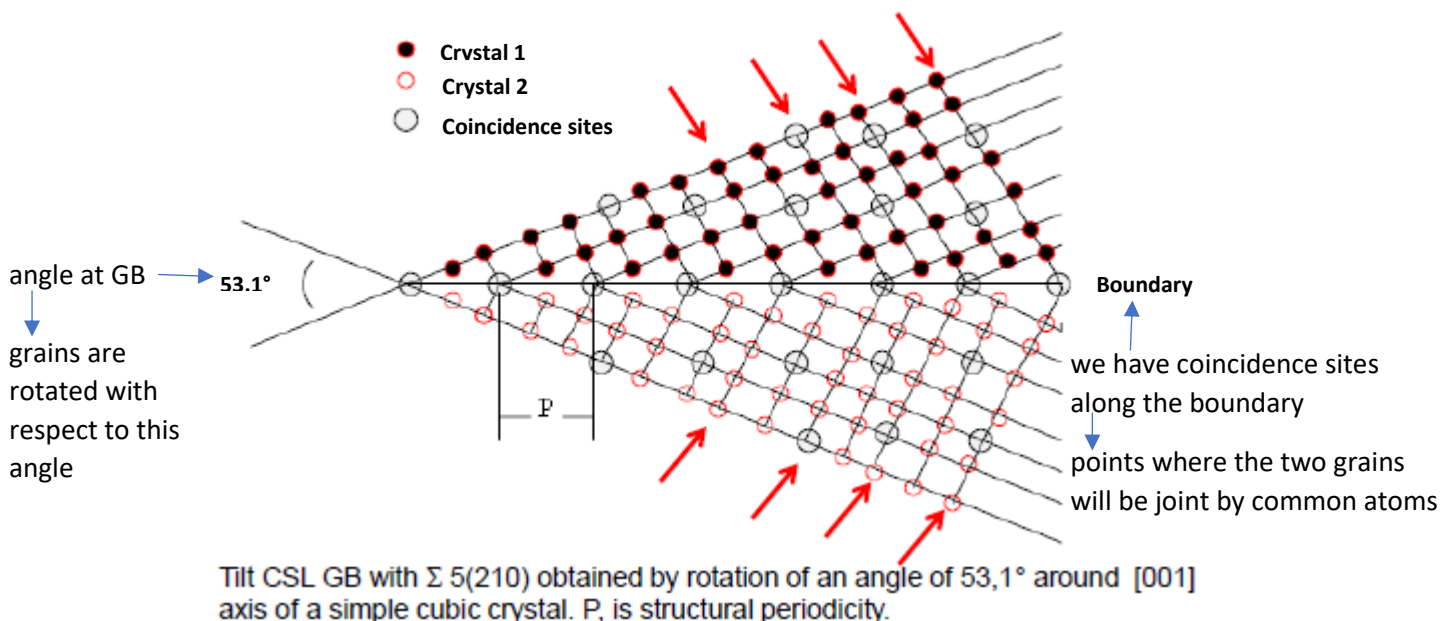
- Ideally the CSL can be visualized as two compenetrating lattices which form a superlattice.
- From each adjacent lattice a common point is selected and by mutually rotating the two lattices the special coincidence orientations are obtained (CSL).
- The intersection of a plane in this lattice and the elimination of the atoms from one side of a crystal of the plane and of the atoms from the other crystals over the opposite side will originate a GB.
- The fraction of coincidence atoms is called **coincidence site density**.

- The **reciprocal** of this density is indicated with  $\Sigma$ , a parameter used to describe a boundary containing the coincidence site.
- e.g. The tilt boundary in figure, being characterized by the directions 210 and 310, has  $\Sigma=5$ .
- The boundaries with  $\Sigma < 5$  and parallel to planes having low CSL boundary index have shown to possess special properties, among other, very low energies and perfect periodicity.

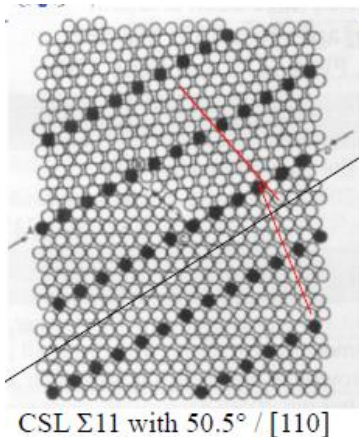
CSL are characterized by:

- Lower GB energy because there are more atoms on grain boundary, more compact like a simple crystal so it has lower energy leading to slower diffusion;
- Higher conductivity because we are eliminating grain boundaries and GBs are source of loss of conductivity (conductivity is higher in single crystal rather than polycrystal);
- Higher creep resistance because we are repressing GBs;
- Higher corrosion resistance because reactivity will be lower;
- No GBs – segregation will not be easy because material will be homogeneous;
- Higher resistance to crack;
- Higher ductility – no GBs, greater flow;
- Lower strength because we do not expect any dislocation to fall into GBs (otherwise dislocations will stop at GBs and the strength will increase);

#### Tilt CSL GBs



#### Coincidence Site Lattice Boundaries: $\Sigma 11$



The production of new polycrystalline alloys with high CSL density will promote significantly improved properties:

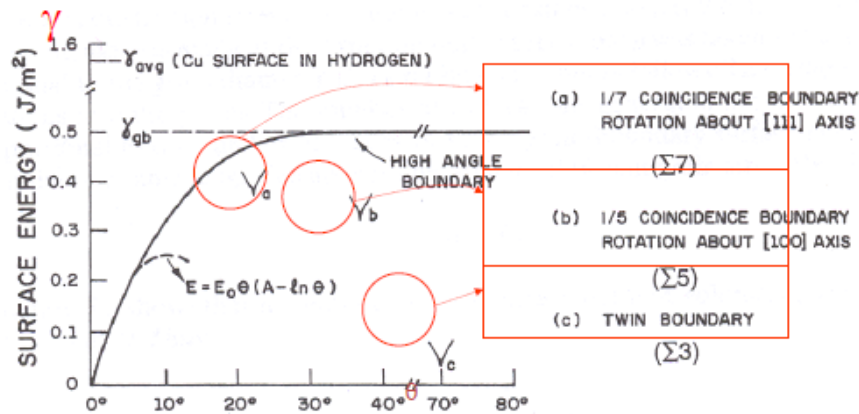
- Lower  $\gamma_{GB}$  in pure metals - Greater diffusivity
- Greater electrical conductivity - Lower susceptibility to segregation
- Greater GB sliding resistance and greater fracture/cavitation resistance
- Greater resistance to localized crack initiation
- Greater resist. to localized corrosion
- Greater GB mobility with specific solutes



← Reduced corrosion sensitivity in austenitic stainless steels



### Surface (GB) Energy as a function of misorientation angle $\theta$ (Cu case)



GB energy  $\rightarrow$  highest energy we have.

The smaller the angle the lower the number of atoms before common sites.

Increase of the surface energy with the increase of the misalignment angle. With a lower angle, we have almost a linear behavior between surface energy and angle.

### Grain Boundary Ledges (GBL) in large-angle GBs

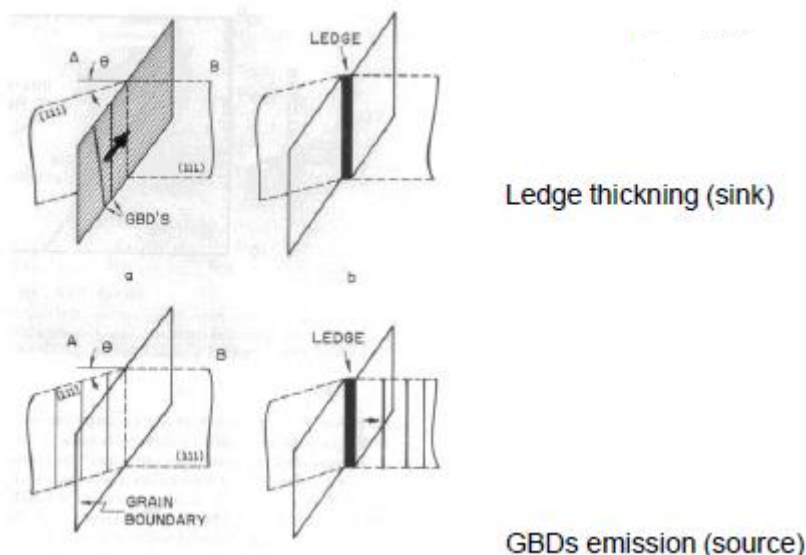
Large-angle GBs have the maximum GB energy and a lower number of common sites.

Experimental observations show that **Grain Boundary Dislocations (GBD)** may form when the misorientation angle does not correspond to the ideal angle required for forming CSL GBs. Dislocations, once inside the grain, move and enter GBs and become GB dislocation. A GBD is locked to the GBs and it is not a common dislocation. It will not be shared by grains, but it will belong to the grain that produces it.

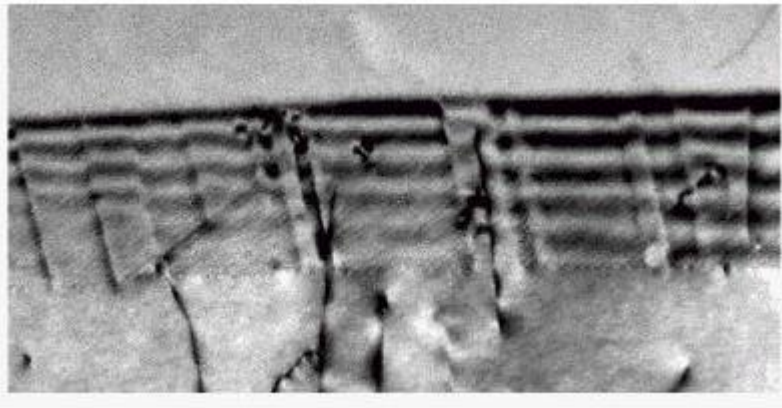
GBDs may cluster into the GB and contribute to the formation **Grain Boundary Ledges (GBLs)**. These GBLs represent an important structural feature of large angle GBs, as dislocations would be able to jump in case of small angles. The density of ledges increases with increasing angle.

The ledges may function as **powerful dislocation sources**, a fundamental factor to explain the mechanical properties in polycrystals. In fact, for plastic deformation to occur, we need dislocations to move. Dislocations start moving from inside the grain and it may end at GB, which may make a dislocation die. We need ledges as a source to make dislocations stronger (another source is the Frank-Read source that we will see later).

Dislocations start moving when under **shear stress**. The higher the shear stress, the higher the amount of dislocations emitted into the grains to create dislocations movement.



### Dislocation grain boundaries in Al



When a dislocation ledge is under Shear stress, it emits dislocations Into the grain → proof for ductility

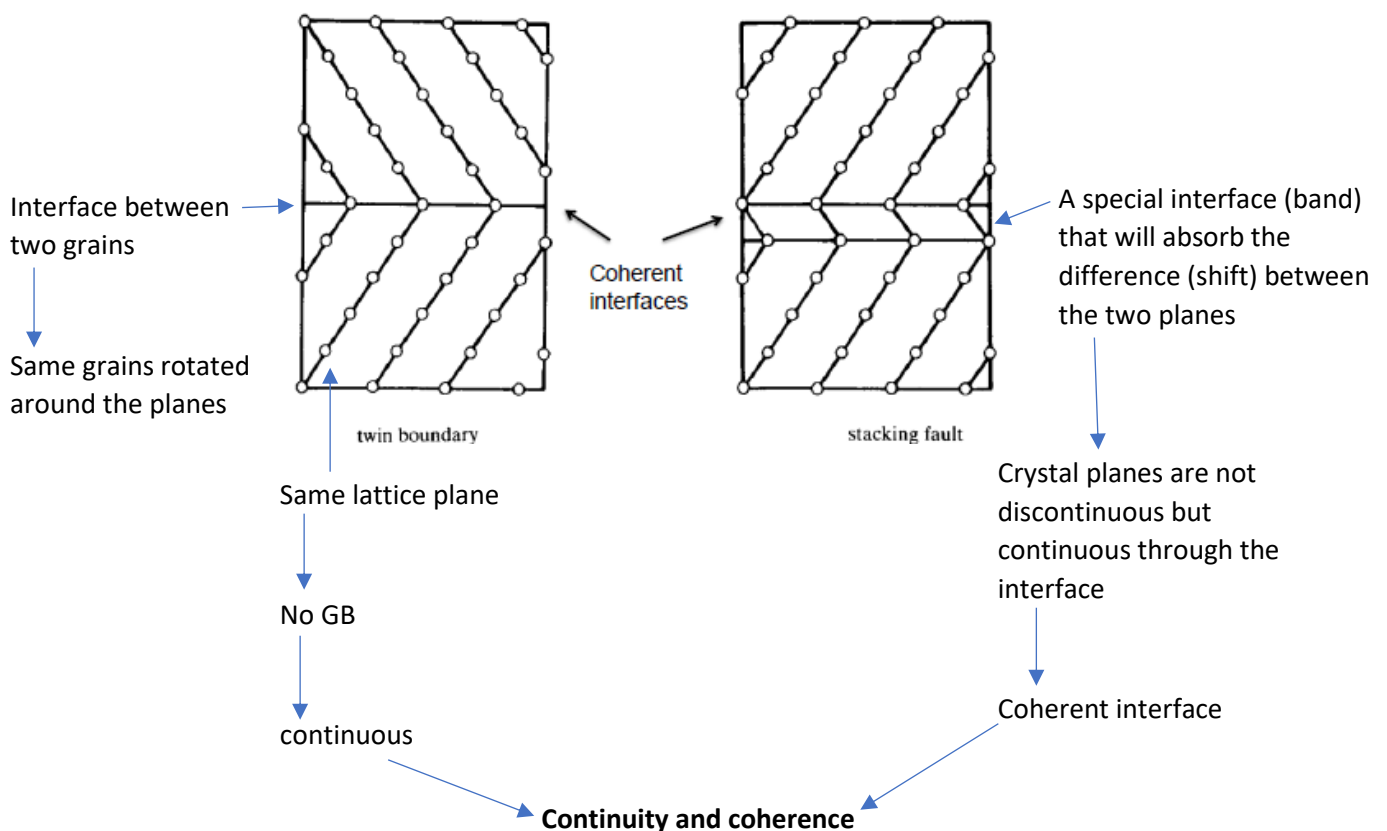
Dislocation (steps) along a grain boundary

### Consequences of GBs

1. They are sources and sinks for dislocations
2. They contribute to mechanical strengthening and change the sliding system → they may deviate the dislocation path
3. They act as sinks for impurities dissolved in the alloy
4. They contribute to the formation of second phases – primary nucleation sites for the growth of second phases
5. They are material portions with no crystallinity

### Packing defects: twin boundary and stacking fault

Two typical 2D coherent defects. The first is named growth-stacking fault or simply twin-boundary, the second is named stacking-fault.





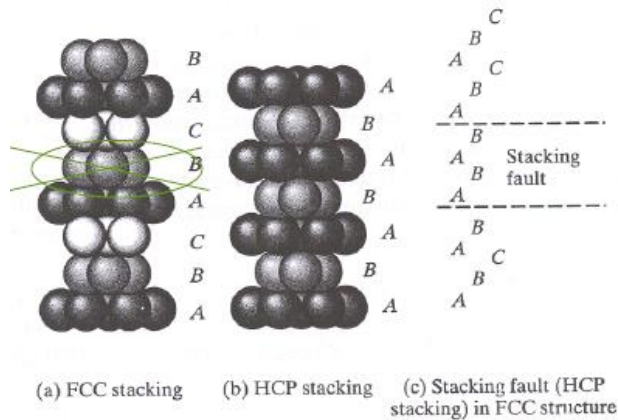
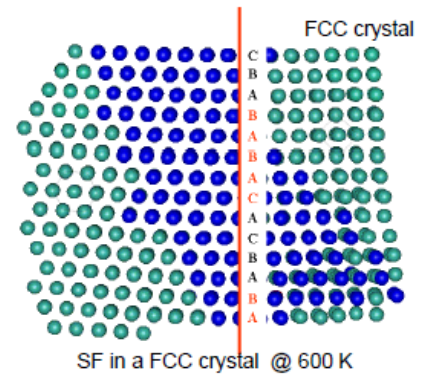
### Stacking fault

It is a region of the crystal which through its interface a part of the crystal is relatively translated with respect to the other of a certain amount being different from  $b$ . It is originated during:

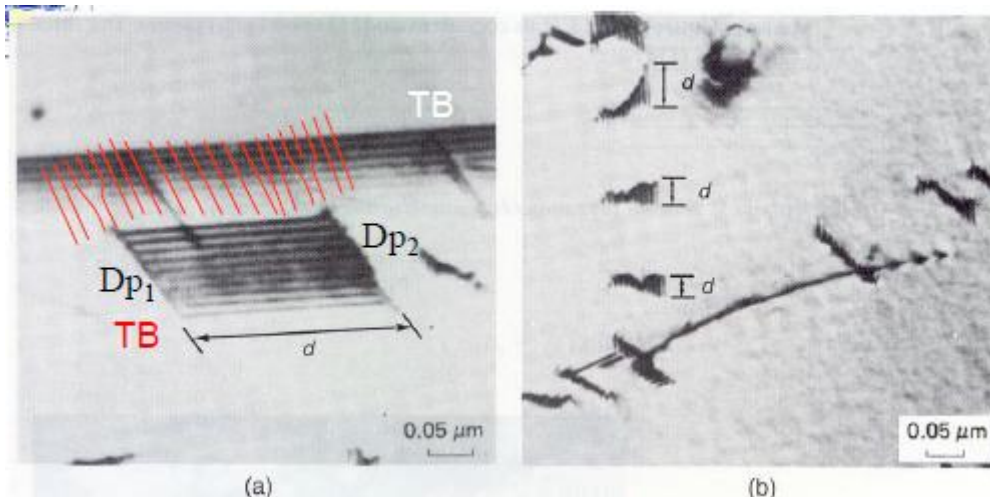
- crystal growth (production of alloys)
- Vacancy/interstitial condensation (Frank loops)
- Plastic deformation, as a consequence of the formation of partial dislocations

When partial dislocation are produced, SF result in FCC as a lattice imperfection due to a mis-packing of closed packed atomic planes whereas the neighboring lattice is unaffected.

The defect is therefore limited to a small region (nanoscale region). Typically, the stacking sequence of planes in a perfect FCC lattice is ABCABCABC... Because of the formation of a SF the stacking sequence becomes ABCABABCABC., with ABAB just the SF region; as the C-atomic plane is missing within the SF region, the stacking sequence becomes that of an HCP in place of a FCC structure.



### Observation of SF with HRTEM

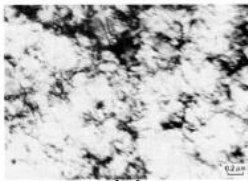


- SF in an AISI 304 austenitic stainless steel (region with small fringes); near the SF a twin boundary (TB) is observed; as SF and TB are characterized by a coherent interface they are visualized with fringes. When observed with electron microscopy. Both defects are discerned by the different contrast.
- perfect dislocations dissociate into partial dislocation thereby producing a train of small SFs.  $d$  is the distance between the partial dislocation pair (also denoted as  $d_s$ )

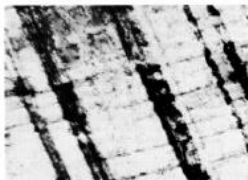
An image that is more likely to be observed on FCC because stacking fault is found on FCC but they don't affect BCC.

### Influence of stacking fault energy ESF on mechanical properties

The stacking fault energy (ESF) is a factor which makes dislocation arrangement quite distant. The lower ESF the higher is the probability of getting stacking force which changes the arrangement of dislocations. ESF changes depending on the alloying element.



(a)



(b)

The way in which dislocations arrange into materials can be different: they arrange as cells (a) or bands (b)

a) >>ESF in a Fe-34%Ni alloy

b) <<ESF in a Fe-15%Cr-15%Ni alloy

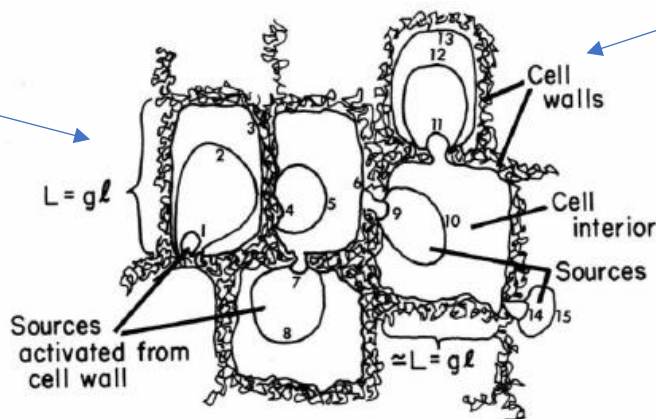
The two alloys have the same total (30%) amount of alloying elements but individually a different content. Both structures are subjected to the same loading condition of 7.5 GPa for 2 microseconds.

SFE sensibly decreases with impurities content (and nature)

### Substructure during cold deformation

The single grain becomes characterized by a number of cells with their own dislocation walls

It's like we have different grains into one grain



Schematic morphology of cellular structure

### Stacking fault Energy

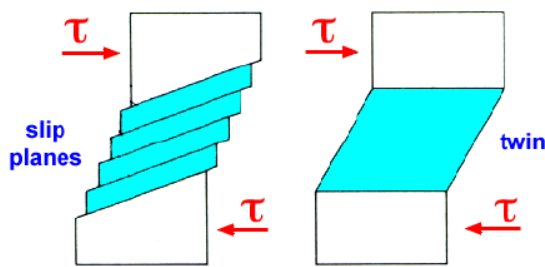
| Metal    | Stacking-fault energy, $\gamma_s$ , J/m <sup>2</sup> | Equilibrium spacing of partial dislocations, $d_s$ , Å |
|----------|------------------------------------------------------|--------------------------------------------------------|
| Silver   | 0.02                                                 | 18.7                                                   |
| Gold     | 0.06                                                 | 5.7                                                    |
| Copper   | 0.07                                                 | 6.8                                                    |
| Aluminum | 0.20                                                 | 1.5                                                    |
| Nickel   | 0.40                                                 | 2.0                                                    |

← Larger for smaller ESF

← High ESF

### Mechanism of plastic deformation

A material can be deformed plastically in any shape and with temperature the material will return without any loss of energy → it will plastically (and slowly) change again to return to the original shape → **Twinning**: opposite of dislocations' plastic deformation because it doesn't require any energy dispersion, while in the case of dislocation (**slip**) we have a dissipative phenomenon because of friction.

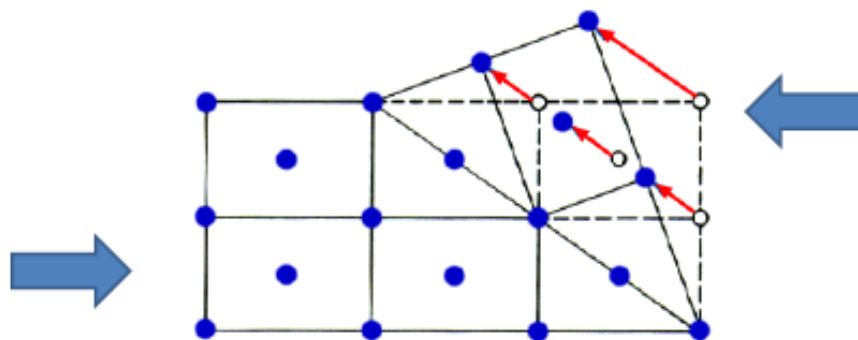


**Slip**  
 Involves all volume of the crystal  
 Slip leaves a series of steps (lines) at the free surface after deformation  
 The lattice direction does not change  
 All atoms on one side of the slip plane move equal distances

**Twinning**  
 Takes part into a small fraction of the crystal volume, the amount of deformation is small  
 Twinning leaves a small but defined twin after deformation  
 The lattice direction changes  
 Atoms move distances proportional to their distance from the twinning plane  
 corrisponde a una frazione della distanza interatomica

As the crystal direction changes, twinning may favor the crystal to undergo further slip.

### Twinning



The imposed shear stress causes a mirror-like rotation of one part of the crystal about the twin plane.

-Twinning, like slip, occurs along well defined planes and directions which depend on the crystal structure (e.g. In BCC crystal twinning is promoted in the  $(112)[111]$  system.

Twins are low energy defects, and their TB have a coherent interface. They are present in so-called shape memory alloys (e.g. NiTi); when deformed original twins disappear; after convenient heating to a specified temperature twins are recovered, and the shape returns to its original state.

BCC cannot twin and cannot exhibit stacking fault!

Tensile test gives the fracture limit of the material.

Twinning increases as alloying increases.

### Slip vs Twinning

Twinning is controlled/promoted by:

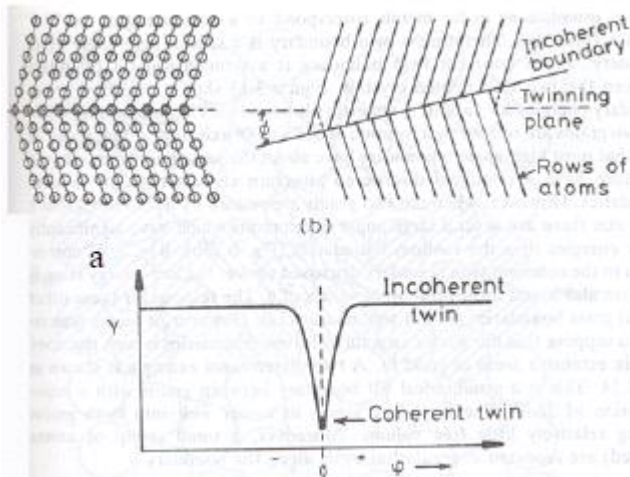
- High strain rate
- Low temperature
- Triaxial stress (material not deformed like in the case of uniaxial stress, but highly constrained. The material will find a way to deform by rotation)
- Increase of alloying
- High ESF
- Fine microstructure

Slip is controlled/promoted by:

- Low strain rate
- High temperature
- Decrease of alloying
- Low ESF
- Coarse microstructure



## Energy and interface Coherence: TBs and SFs

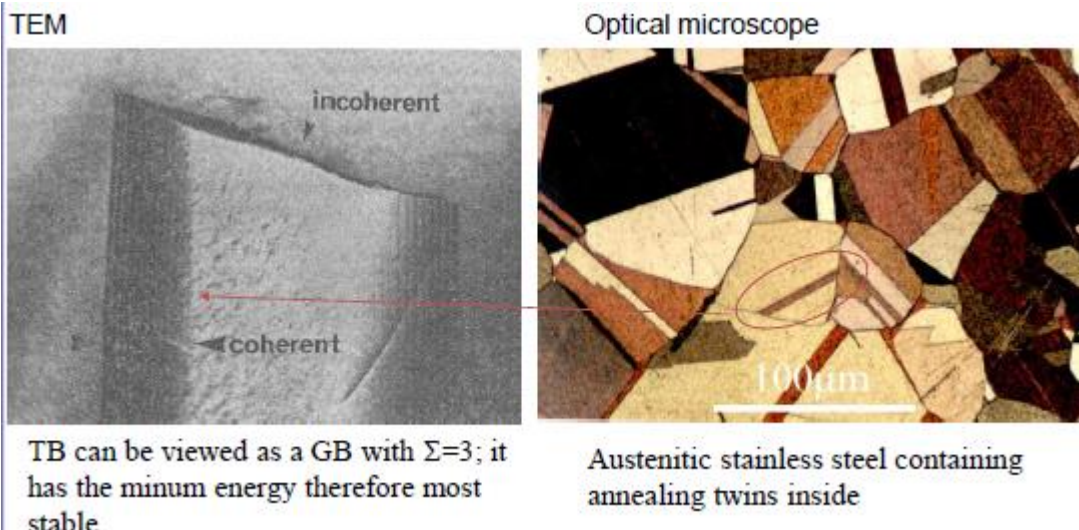


The origin of a TB angle  $\phi$  makes the TB interface incoherent

When the TB angle is zero, the TB becomes coherent and its energy is minimum

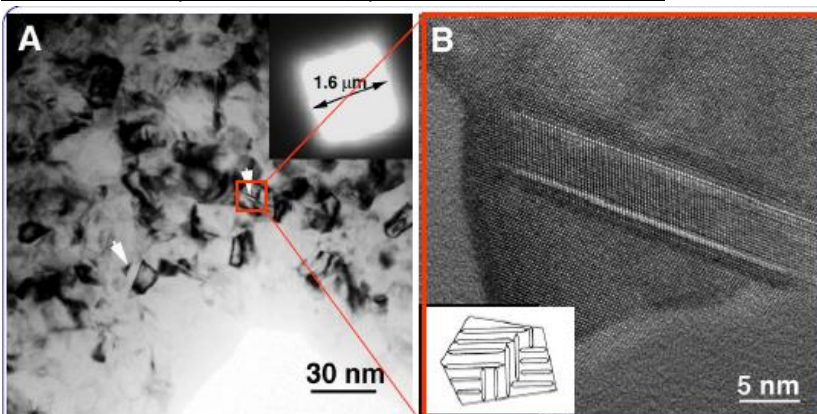
a) Coherent Twin boundary; b) Incoherent Twin boundary; c) energy of a twin boundary in function of TB orientation.

## Twin boundaries



Inside the grains we have straight bands- twins. Twin is formed after annealing at this level.

## Deformation by twins in nAl (by M. Chen, Science 2003)



By time it was an exceptional finding since in Aluminum ESF is very high. Observable at a very low grain size microstructure. Grains are so small that the material prefers to rotate rather than to slip

TEM: twin generated around a hardness indentation. HRTEM: high magnification of the twin observed previously in fig A. Its morphology suggests its origin from plastic deformation.



### Energetics of 2D superficial defects

| mJ/m <sup>-2</sup>    | Al  | Cu  | Pt   | Fe  |
|-----------------------|-----|-----|------|-----|
| Stacking fault energy | 200 | 75  | 95   | -   |
| TB energy             | 120 | 45  | 195  | 190 |
| GB energy             | 625 | 645 | 1000 | 780 |

Grain boundaries energies are far much larger than those of other defects and thus effective to retain dislocations.

### 3D lattice defects

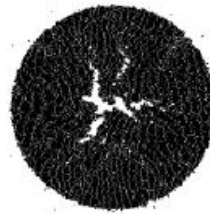
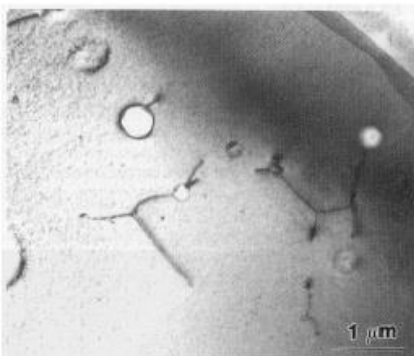
Microvoids, microcracks, triple point, quadruple points, etc.

Fracture propagates more likely between grains → intercrystalline fracture

The larger the grain, the faster the crack because it must change the direction so often in order to break the final material

Susceptibility for fracture is lower with lower grains size

### Microvoids and microcracks

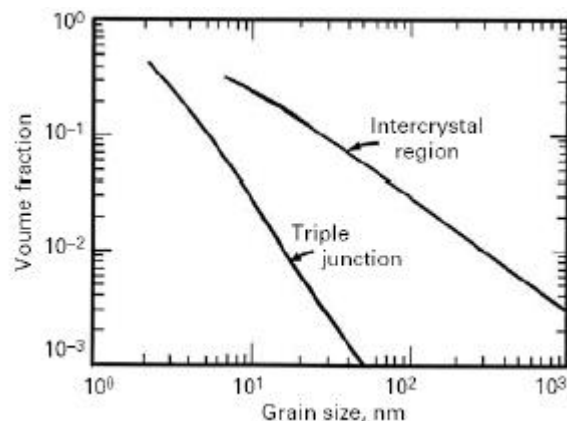
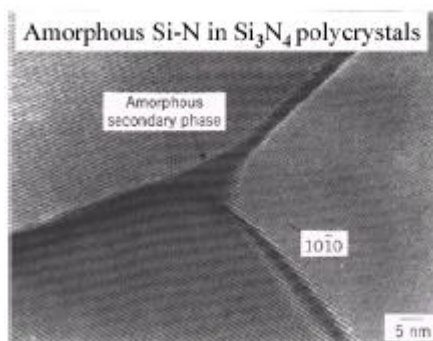


Clustering of nanocracks

Microvoids in TiC: they constitute an obstacle for dislocations (Meyers)

We start from a pore which will be a crack. When the defect becomes greater they become very dramatic for the stability of material

### Triple points: three-grain junction



Triple- or quadruple junctions are special sites where 3 or 4 grains join. They are favorably encountered in nanocrystalline materials. The number of triple junctions have a large influence on the mechanical properties of polycrystalline materials.

The finer the grain size, the greater is the amount of triple points.

Triple points contribute to diffusion. They change the direction of atoms during diffusion in a very quick way. Small grains lead to more triple points which implies faster diffusion

When the material is nanostructured, it becomes less ductile, since the probability to have dislocations is low? Wrong! The material can still flow with GB sliding or by twinning. With the decreasing of grain size, the material becomes more resistant.

### Summary of defects' effect on properties

| Defect        | type              | Improved Materials Properties                                                                                                                                | Adversely affected Materials Properties                                                                                                     |
|---------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Point Defect  | Vacancy<br>$f(T)$ | <ul style="list-style-type: none"> <li>– Diffusivity</li> <li>– Color Centers</li> <li>– Ionic Conductivity</li> </ul>                                       | <ul style="list-style-type: none"> <li>– Electron mobility</li> <li>– Carrier Lifetime</li> </ul>                                           |
|               | Substitutional    | <ul style="list-style-type: none"> <li>– Conductivity (dopant)</li> <li>– Strength (hardness)</li> <li>– Characteristic T (like <math>T_M</math>)</li> </ul> | <ul style="list-style-type: none"> <li>– Conductivity (impurities)</li> <li>– Ductility</li> <li>– Characteristic T</li> </ul>              |
|               | Interstitial      | <ul style="list-style-type: none"> <li>– Strength</li> <li>– Characteristic T</li> <li>– Electrical Properties</li> </ul>                                    | <ul style="list-style-type: none"> <li>– Ductility</li> <li>– Characteristic T</li> <li>– Electrical Properties</li> </ul>                  |
| Line Defect   | Dislocation       | <ul style="list-style-type: none"> <li>– Ductility (Malleability)</li> <li>– Strength (at high density)</li> </ul>                                           | <ul style="list-style-type: none"> <li>– Strength</li> <li>– Yield Stress</li> <li>– Optical Properties</li> <li>– Lasing Action</li> </ul> |
| Planar Defect | Grain Boundaries  | <ul style="list-style-type: none"> <li>– Strength</li> <li>– Electrical Properties</li> <li>– Magnetic Properties</li> </ul>                                 | <ul style="list-style-type: none"> <li>– Creep</li> <li>– Electrical Properties</li> <li>– Magnetic Properties</li> </ul>                   |

## Lecture 6

### Strengthening of alloys

Metals, as they are produced, are highly ductile. We want them to be strong and in order to do that we need to control the microstructure to get a better compromise between application properties. We exploit different factors to increase strength.

Strengthening of metals and alloys:

- Strengthening by s.s.s (substitutional solid solution) and i.s.s (interstitial solid solution)
- Work hardening
- Strengthening by grain refinement
- Strengthening by precipitation (aging)
- Strengthening by dispersion of hard second phase
- Strengthening by hard phase (e.g. Martensite)
- Strengthening by fiber orientation (composites, texture)

The total strengthening is given by the summation of all contributions:

$$\tau = \tau_{lattice} + \tau_{s.s} + \tau_{prec} + \tau_{strain-hard} + \tau_{GB} + \tau_{hard-phase} + \tau_{fiber}$$

$\tau_{lattice}$ : pure lattice FCC, BCC, HCP, make dislocations move in a different way. Any crystal lattice applies its own resistance to dislocations motion depending on its structure. the crystal lattice introduces its own strengthening.

$$\tau_{int} = \frac{2G}{1-\nu} \exp\left(\frac{2\pi a}{(1-\nu)b}\right)$$

a: distance between sliding planes

b: distance between atoms along the sliding direction

G: tangential modulus (elastic modulus depending on metal.)

$\nu$ : poisson coefficient

$\tau_{int}$ : tangential stress → because the dislocations are related to shear stress, I can make dislocations move in the system only when I apply shear stress → difficulty in making dislocations move will translate into strengthening the resistance of the material.

#### 1) Strengthening by solid solution

Alloys are stronger than pure metals.

Substitutional and interstitial alloys cause lattice distortions that interfere with the strain field created by the dislocations, by hindering their motion.

Impurities tend to diffuse and segregate around the dislocations core in order to find more suitable sites which better fit with their dimensions. This will reduce the overall strain energy and at the same time will “anchor” impurities to dislocations. They make dislocations slow down which leads to strengthening. Dislocations motion → material is flowing plastically. If the material is getting stronger, dislocations are slowing down which makes it difficult for the material to flow. When dislocations stop moving and we add shear stress to promote dislocation motion, we have fracture when reaching UTS.

The dislocation collects impurities and pulls them to higher-energy lattice regions having higher strain, where other dislocations are located. The strain induced by the latter is not compensated anymore by impurities (metastable configuration); a partial relaxation of lattice strain is observed.



By increasing the solute content, the lattice strain will increase as well as their hindering. Hence, the strengthening effect increases.

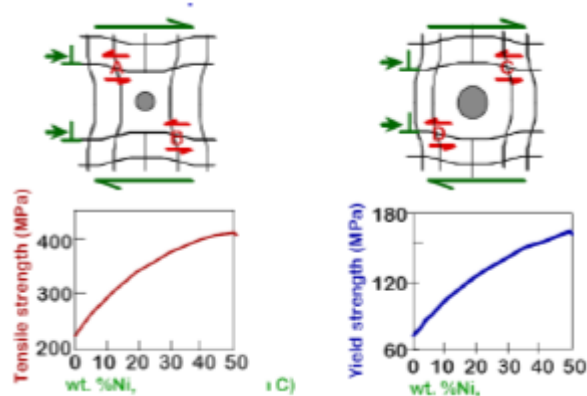
As atoms of solute “b” are introduced to atoms of an alloy “a”, the size of the diameter of “b” will change the structure.

$D_b < D_a$  - higher strength

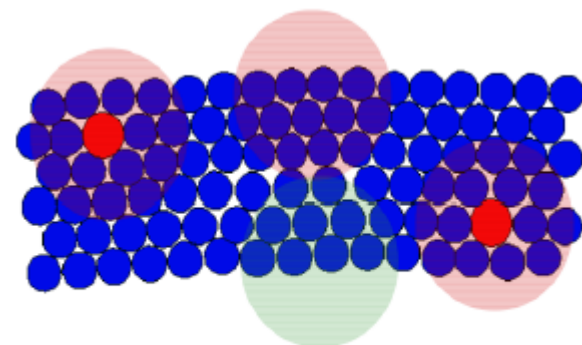
$D_b > D_a$  - higher strength

What is important in strengthening the material is the difference in the size of atoms and not which is smaller or larger. The higher the difference, the higher the strength.

The smaller the atom, the higher is the interatomic force and therefore the higher the strengthening.



As the amount of Ni increases, the tensile and the yield strength will increase. When we introduce foreign atoms into a solid solution, a strength field will take place. Such strength field has a bipolar nature: compressive and tensile.

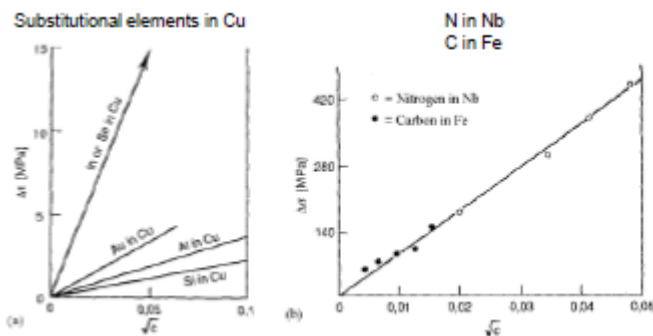


Add a few atoms of size different from the basic metal

compression tension

Lattice changes when a new atom is added.

There is a change of the overall stress and strain state of an alloy → strengthening of alloys properties uniformly due to the generation of compressive and tensile regions.

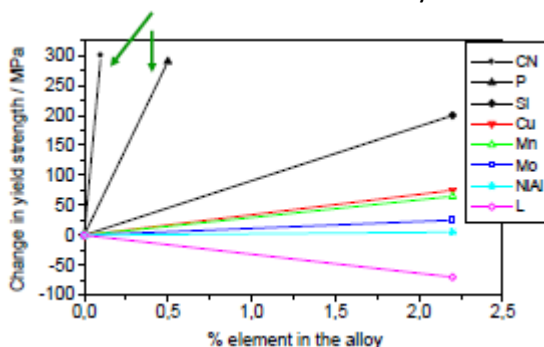


The strengthening effect increases with the square root of the concentration of solute, be it substitutional or interstitial.

← Strengthening is quite significant. Even a small amount of In or Sn produces a very high strengthening into Copper because the diameter difference is very large.

Once I am able to fill the lattice with the new atoms and I get distortions with compressive and tensile fields, the alloy will be deformed as we do with a wire so I'm moving dislocations inside such material. Dislocations will move inside a region affected by the strength field introduced by substitutional solutes.

Dislocations will not move easily in the compressive field while they will be free to move in the tensile field.

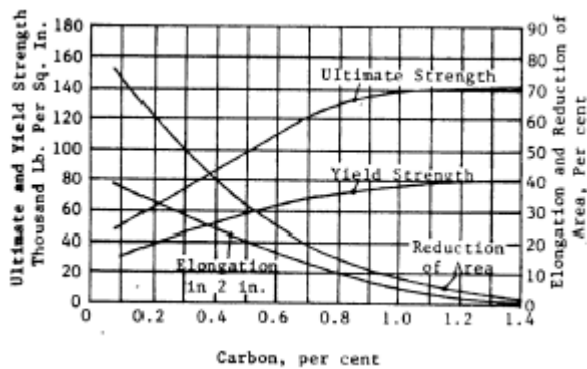


Therefore, not only the difference in size between atoms affects the distortion around the atom site but also the strength field very much distant from these atoms will be effective.

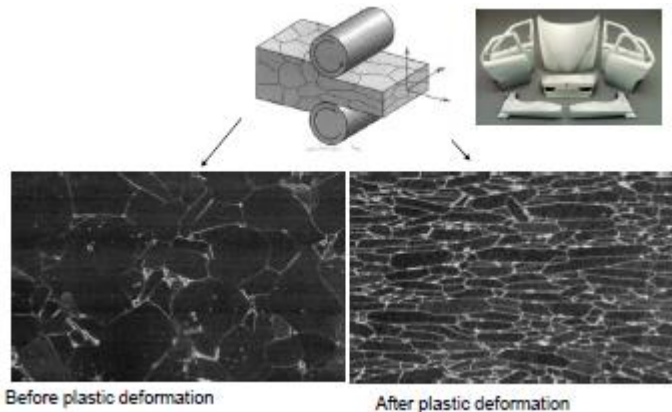
Strengthening by interstitials is one of the most effective strengthening methods.

Regardless of the type of atom (interstitials and substitutional), the strengthening is always linear with respect to  $\sqrt{c}$  which must be viewed as a separation distance between atoms.

## Effect of C in steels



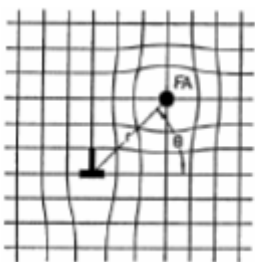
## 2) Strengthening by work hardening



Rolling process: the material is pressed between two counter-rotating rolls and it is squeezed but also advanced along rolling direction (grain flattened and elongated) → introduction of a significant amount of dislocations. We have also precipitates at GBs in order to keep grains as small as possible.

## Strengthening between dislocations and interstitials

If we have an interstitial in a lattice it will distort the lattice. If the material is cold worked (like rolling) dislocations will generate and will move across the lattice. Whenever the dislocations will feel distortion, they will slow down (the material will be strengthened).

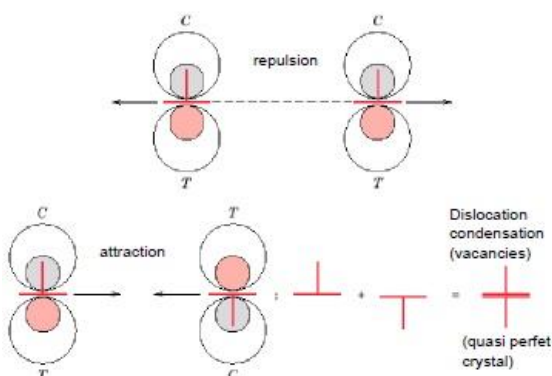


$$\Delta\tau \propto \sqrt{c}$$

- Paraelastic interaction: very large lattice to interstitial atomic radii ratio
- Interaction with the strain field induced by the dislocation and that of the interstitial

For low strain rate values (low dislocation speed), impurities can follow dislocations by diffusion.

## Interaction between dislocations



When dislocations move on the same slip plane with the same side, we have repulsion and the formation of interstitials. When they move with opposite sides, we have attraction and the formation of vacancies.

At room temperature, dislocations are forced to move on the same plane.

### Work hardening: Orowan Equation

Orowan was a scientist who explained the mechanism of dislocations while moving into lattices and the relationship between shear strain  $\gamma$  and the dislocations motion.

A single dislocation that moves along the slide plane (area A) will determine a strain  $\gamma = b/d$  ( $b$  is the Burgers' vector and  $d$  is the dislocation length)

- $N_d$  = n. of moving dislocation to a specified distance  $dx$ , causes a differential sliding

$$d\gamma = N_d \left( \frac{dx}{A} \right) \frac{b}{d}$$

- By definition density of mobile dislocations is:  $\rho = N_d / A d$
- Amount of slip  $d\gamma = \rho b dx$

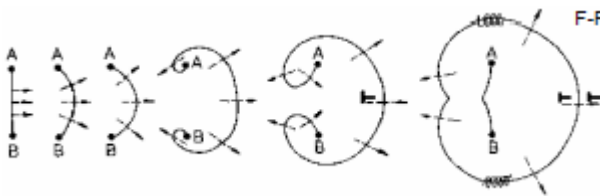
$$d\gamma/dt = \rho b v$$

$v$  = average velocity of dislocation in  $dx$  space

The strain rate in the material is proportional to the dislocation density, to the Burgers vector and to the average velocity of dislocation. The higher the value of  $\rho$ , the higher the strain rate. The higher the  $v$  the higher the strain rate

### Strengthening due to F-R (Frank-Read source)

It's linked to precipitation strengthening. The strengthening is caused by the interaction between a dislocation and two particles.



**F-R Source** Given two particles A and B, the dislocation pressed by shear stress will interfere with them and instead of shearing them (because they are quite large) it bends like a wire. As the stress increases the wire will bend more and more in a curly shape  $\rightarrow$  the dislocation will expand and

will turn back to the particles making such a curly shape. At the end, the two ends will bond by making one ring. The original dislocation will be reformed and that's why it's named a source: it generates many rings and dislocations. The rings once formed around the particles are fixed and can't bend and don't move.

The curvature radius  $r$  depends on the resolved shear stress  $\tau$ . There establishes a critical bending when  $r = L/2$  ( $L = AB$ ), from which:

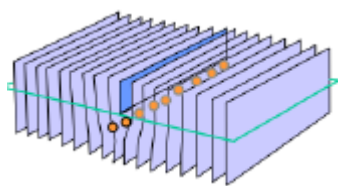
$$\tau = G b / L$$

$L$  is the dislocation length and  $G$  is an elastic measure of material.

The next step of the process is represented by a generation of dislocation rings condensation of dislocations segments having the same  $b$  but different direction.

### Interstitial solutes - dislocation "pinning"

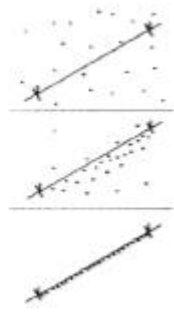
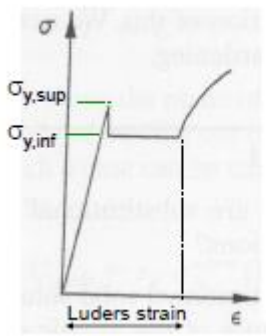
Dislocation is pinned by an array of atoms attached to them, interfering with their movement. The lower the stress applied, the higher the pinning.



Two features of interstitial solutes in BCC metals make them powerful hardening agents:

- They have very large stress fields around them, so they can lower the energy by "sitting" at dislocation cores
- They can diffuse rapidly even at room temperature, so they can "find" the dislocations in hours or days.

### Cottrell atmosphere and Luders' band



At room T, C and N atoms are relatively free to move even over long distances. They segregate nearby superficial dislocations to reduce the system energy. A larger stress is required to contrast and let the dislocation free during their motion.

$\sigma_y$  increases ( $\sigma_{y,sup.}$ ).

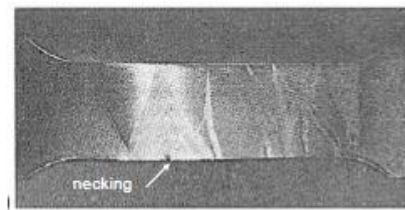
The obtained configuration is unstable. When the strain is equal to the Luders' deformation, a sudden disanchoring occurs and the deformation undergoes an abrupt increase though in the absence of an applied stress ( $\sigma_{y,inf.}$ )

In other words, once a dislocation is pinned, a small extra force is required to unpin the dislocation prior to yielding, producing an observed upper yield point in a stress-strain graph. After unpinning, dislocations are free to move in the crystal, which results in a subsequent lower yield point, and the material will deform in a more plastic manner.

For low strain rates (low dislocation speeds), impurities can proceed to follow the dislocation motion by diffusion.

When the small amount of particles is present, strengthening is not significant. The dislocation line is collecting all atoms it's finding on its way. Mild steel- a small amount of interstitial (Cn). Stress depressed when dislocations are free to move.

### Luders' bands in a tensile test sample



localized stress

- **Stage 1:** strong localization of strain, i.e. at the surface the dislocation become more free to move disanchoring from Cottrell's atmosphere and the plastic deformation will restart in the form of Luders's band → orientation is 45° with respect to loading direction (Taylor factor, Mmax)
- **Stage 2-3:** The formation of Luders' band is accompanied by necking phenomena (formation of unstable necks); → fracture phenomena or
- **Stage 4:** Work hardening prevents fracture: stress intensification to the neck tends Luders bands propagate from the necking region throughout the sample

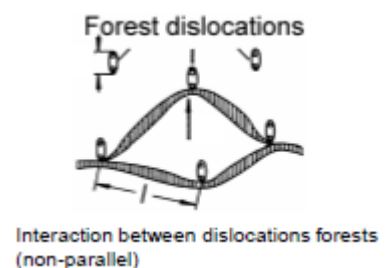
### Strengthening by dislocation forests: work hardening

Interaction of dislocations results into the formation of dislocation forests. Dislocations interact and form defects like vacancies, but they strengthen the material because they cannot move further, they slow down.

The stress applied to a dislocation to overcome other parallel dislocations is given by:

$$\tau \approx \alpha G b \sqrt{\rho}$$

Distance between dislocations:  $d \approx \rho^{-1/2}$ ,  $\alpha = 0.1 \dots 1$



### Strengthening in polycrystals

Work hardening (is the increases of dislocation density which) accompanies the increase in dislocation density which, in turn, makes enhance the plastic deformation.

Ductile metals strengthen when they are plastically deformed at temperature much lower than the melting temperature.

The degree of cold working (%CW) is often used to express the work hardening of a metal or alloy (or ductility)



$$\%CW = \left( \frac{D_o - D_f}{D_o} \right) \times 100$$

$$\%CW = \left( \frac{A_o - A_f}{A_o} \right) \times 100$$

where  $A_o$  is the initial cross-section area,  $A_f$  is the cross-section area after deformation ( $D_o$  = initial diameter,  $D_f$  = final diameter).

### Plastic deformation in polycrystals

We have a polycrystal with many grains which are oriented in a different way, where the sliding direction changes from crystal to crystal. The dislocation will be created inside the grains and after moving it will interact with GBS. If the angle of orientation of grains is too high, dislocations cannot cross GBS; they will pile up at GBS, they will stop there, increasing the dislocation density. If GBS angle is low, dislocations can cross. While in simple crystals dislocations may originate and move freely and their slowing down may be affected only by their interaction, in polycrystals we have an additional defect, GBS, which opposes the motion of dislocations leading to the increase of strength.

Therefore, polycrystalline metals are much stronger than single crystal metals (except for perfect monocrystals named **Whiskers**).

Whiskers are very hard, elastic, high strength, they don't allow motion of dislocation, they don't need any strengthening mechanism.

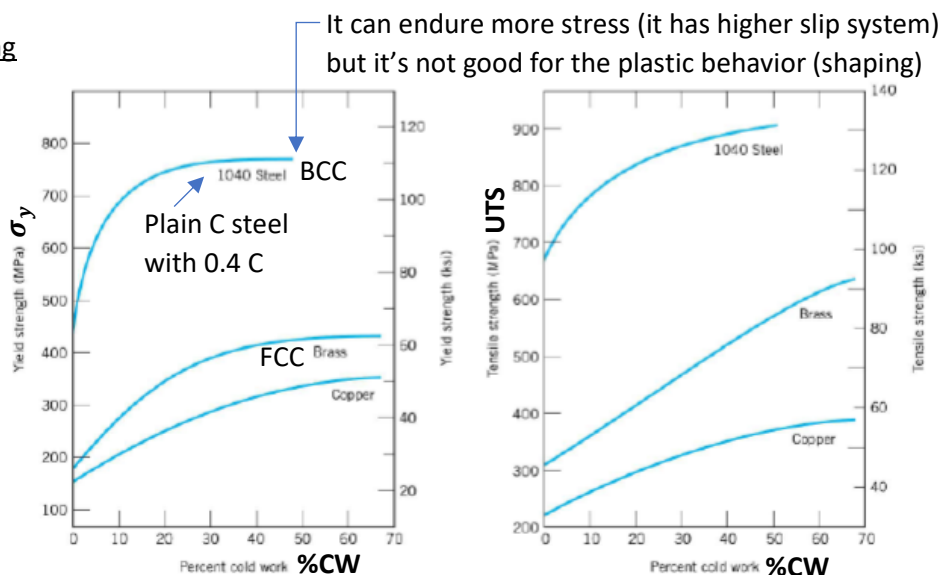


The orientation of individual grains into a polycrystal with respect to the load direction is random. The dislocation movement occurs along sliding systems having favorable orientation (i.e. that is the largest resolved tangential stress).

After high plastic deformation, crowding of the lines are present and the change of orientation angle in order to fit higher dislocation inside. Every grain has its own slip system. Dislocations will move along their own slip system and when they reach GBS they will interact with other slip systems near neighboring grain. This is the case of Cu, where twinning is possible, grains may rotate slightly.

The lines in the picture represent the dislocation bands.

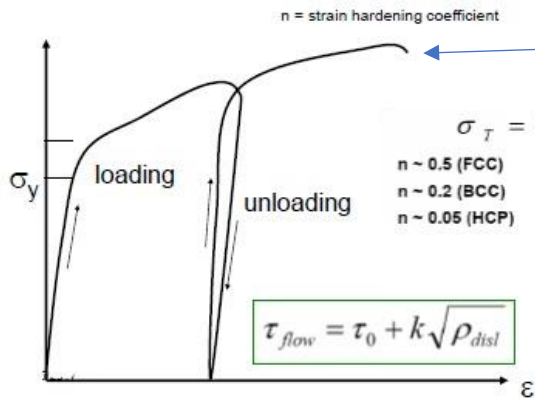
### Work hardening



- Interaction among dislocations => regions in which dislocations are piled-up
- Requires larger deformation to make dislocations restart moving
- Larger strain => strain hardening
- Each increment of deformation requires an increase in applied stress e.g. strain around a print left after a hardness indentation

Grains may re-orient depending on the level of stress applied (grains are more work hardened). Cu alloys can be severely work hardened because they combine twinning and slip to make very intensive work hardening. With the increase of %CW, the strength will increase. In the case of steel, when exceeding 25%, we have saturation (no more strengthening).

### Strain hardening



As the load increases, we have further work hardening up to the work UTS.

The Ludwik- Holloman equation describes only the plastic part of the tensile curve. The larger is n, the larger is the work hardening.

Work hardening is a combination of two effects: straining and strengthening. They should be both present for the material to be work hardened.

Tensile test → After we reach a certain level of work hardening we stop the test, we unload and then reload → parallel lines → the elastic modulus of the steel does not change during the test, only the yield changed (it's like we are working with a different material).

Because unloading and reloading again, the material has acquired additional work hardening.

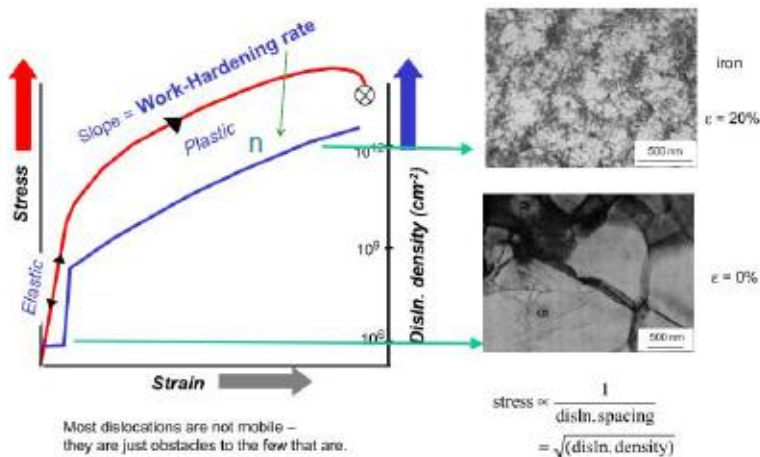
When the material is shaped by any process, it will increase in every point different yield stress and whenever in some point the material will reach the limit UTS after that, the material is unstable and will just break, the process is out of control. We need to control the stress level in every point of the material under shaping in order to ensure higher stability and strength of the component.

### Ludwik-Holloman constants for various alloys

|                                 | K, Mpa | n    |
|---------------------------------|--------|------|
| Aluminum, 1100-0                | 180    | 0.2  |
| 2024-T4                         | 690    | 0.16 |
| 5052-0                          | 210    | 0.13 |
| 6061-0                          | 205    | 0.2  |
| 6061-T6                         | 410    | 0.05 |
| 7075-0                          | 400    | 0.17 |
| Brass, 60-39-1 Pb, annealed     | 800    | 0.33 |
| 70-30, annealed                 | 895    | 0.49 |
| 85-15, cold-rolled              | 580    | 0.34 |
| Bronze (phosphor), annealed     | 720    | 0.46 |
| Cobalt-base alloy, heat-treated | 2070   | 0.5  |
| Copper, annealed                | 315    | 0.54 |
| Molybdenum, annealed            | 725    | 0.13 |
| Steel, low-carbon annealed      | 530    | 0.26 |
| 1045 hot-rolled                 | 965    | 0.14 |
| 1112 annealed                   | 760    | 0.19 |
| 1112 cold-rolled                | 760    | 0.08 |
| 4135 annealed                   | 1015   | 0.17 |
| 4135 cold-rolled                | 1100   | 0.14 |
| 4340 annealed                   | 640    | 0.15 |
| 17-4 P-H annealed               | 1200   | 0.05 |
| 52100 annealed                  | 1450   | 0.07 |
| 302 stainless, annealed         | 1300   | 0.3  |
| 304 stainless, annealed         | 1275   | 0.45 |
| 410 stainless, annealed         | 960    | 0.1  |

Cobalt-base alloy, heat treated, has a lot of strengthening. However, we have to avoid cobalt since it's very toxic.

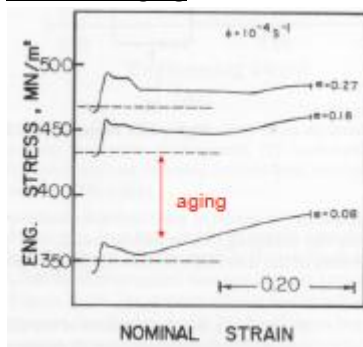
## Work hardening



$n$  will be constant in case the alloy follows Ludwick-Holloman law. The slope represents the derivative of stress with respect to strain. The closer the dislocation the larger the stress/strength ratio. Dislocations will accumulate on Gbs when stress increases. The core of the grain will be affected by dislocations as well.

When polycrystals are activated by dislocations, first to start slipping are Gbs → micro-yielding.

## 1b Strain Aging



Strain aging is the consequence of strain hardening in the materials.

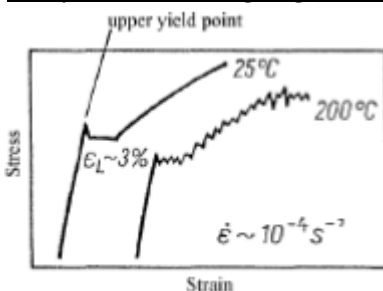
← Stress-strain curves after tensile test applied to three samples and 3h interruption (aging); then reloading → reappearance of  $\sigma_y$  phenomena seen after an interruption.

Initially the alloy does not show any significant yielding after preloading. After prolonged aging impurities segregations do occur resulting in a remarkable yielding point.

Strain aging is mostly seen in mild steels (which are characterized by a low carbon content and the appearance of low and upper  $\sigma_y$  → consequence:

Luders' band → instabilities, undesired when the material undergoes shaping).

## 1c Dynamic Strain ageing (Portevin-Le Chatelier effect)

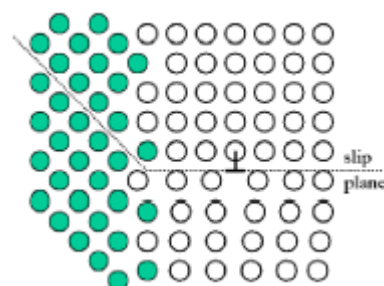


Two-point yielding  $\sigma_y$ , sup/ $\sigma_y$ , inf in mild steels

The steel is strained while heated at low (200 °C) → impurities segregation is accelerated → segregation hinders the dislocation motion.

The observed tensile test curves show a typical saw-shape type curve due to alternating anchoring/disanchoring of dislocations to interstitials (Cottrell's atmosphere).

## 3) Strengthening by grain boundary refinement



GBs impose an obstacle to moving dislocation as they represent either a discontinuity or a misorientation boundary between sliding planes.

Small angle GBs are very effective to hindering dislocation motion.

Large-angle GBs hinder dislocations motion and enhance the mechanical strength of the materials. An intensification of stress at the ends of a sliding plane may work as a trigger to new dislocations in adjacent grains. Hindering makes it difficult for dislocations to move.

GB refinement → possibility of making grain size smaller, since the smaller the grain size the higher the strength; it's the most effective to increase strength and toughness (only possibility).

Interaction of dislocation with Gbs → small angle → dislocations may continue to move into the grain.

GBs behave as a barrier to slip. This barrier imposes a mechanical “resistance” by increasing GBs’ angle. With the increase of dislocations at Gbs the orientation of both grains will change by increasing the Gb angle. If the grain is small, the dislocation displacement is small → dislocation will reach the barrier very quickly, which leads to higher strengthening.

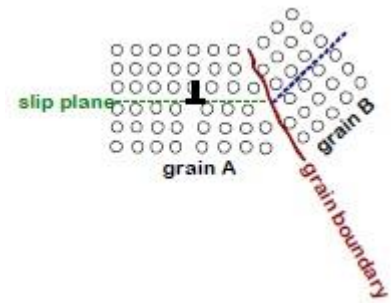
The smaller the grain size, the larger the barrier to slip.

Hall-Petch equation:

$$\sigma_{yield} = \sigma_o + k_y d^{-1/2}$$

$\sigma_o$ : stress measured in the absence of dislocation;  $d$ : diameter of the grain.

$\sqrt{d}$  represents the distance between the dislocation source (center of grain) and the location of Gb; the smaller  $\sqrt{d}$ , the higher  $\sigma_{yield}$ .



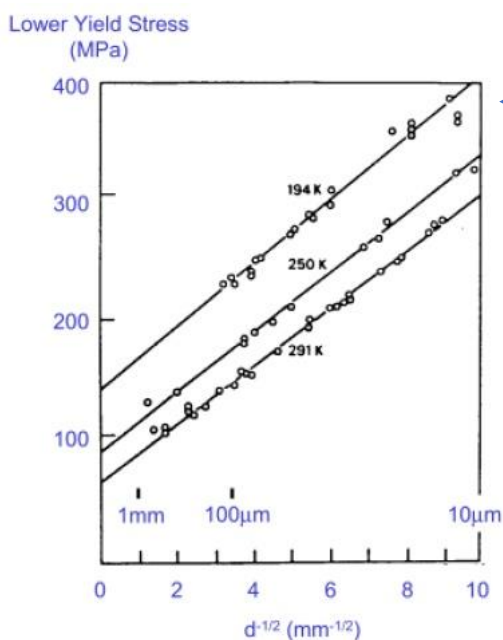
The size of the grains significantly influences the mechanical properties.

At room temperature, coarse grains exhibit low strength, hardness and higher ductility (slow cooling on heat treatment → coarse grains).

To increase mechanical strength  $R_{mech}$  through grain refinement, it is required to:

- heat/cool rapidly → it inhibits grains from becoming larger
- promote pinning of GBs (by allowing hard second phases at GBs) → we introduce solute → at GBs we have precipitates which prevent grains from expanding even at high T and longtime heat treatment
- strain harden, recrystallize-and-cool rapidly (quench) → a material can recrystallize by promoting work hardening and putting the material at recrystallization temperature (40%- 70% of melting T) of the alloy. The material, after work hardening, will have very elongated and distorted grains, with a high concentration of dislocations inside. Most energy from cold working is accumulated at GBs. At recrystallization temperature, recrystallization occurs, new grains are formed by replacing the previous ones. If we don't stop recrystallization, grains will be longer and bigger. Longer grain will capture smaller ones (cannibalism). If the process continues, only one big grain will remain.

Strengthening by GB refinement is the only strengthening method which increases both mechanical strength and tenacity/resilience.



← Effect of grain size on yield strength of mild steel.

Slope of the line → K constant

As temperature increases, the material has a lower strength.

Grain decreases as  $1/\sqrt{d}$  increase.

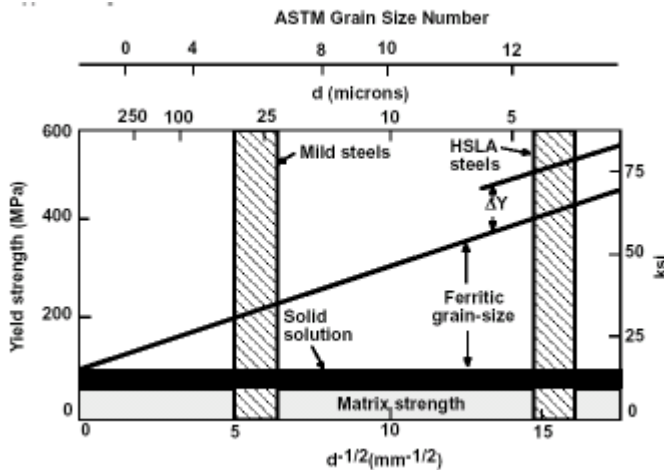
$\sigma_y$  increases with grain becoming smaller.

Note:

- range of grain sizes – typical for an engineering alloy
- (martensite may have interfaces spaced by only a few 10's of mn)
- (in pearlite, interfaces may be a few  $\mu m$  apart)
- ~3x range in yield strength over this grain size range
- Variation of strength with temperature
- Relatively high  $\sigma_y$  due to strengthening mechanisms other than grain size.



## Strengthening in HSLA steels <<%C



**HSLA** = High Strength Low Alloy steels usually made by precipitation hardening and used for middle strength sheet components employed in automotive and civil engineering. They are micro alloys with very small grains → strengthening methods are maximized.

The larger the grain size number, the finer the grain. As the grain becomes smaller, the strengthening is linear.

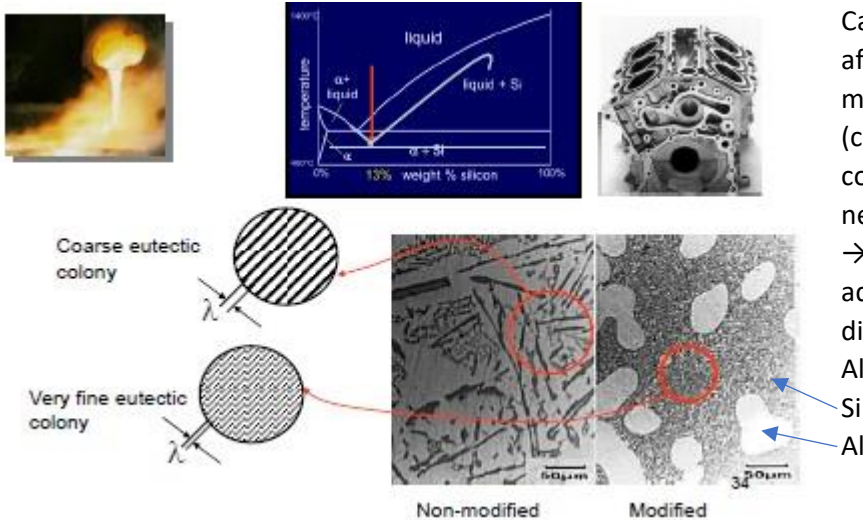
## Hall-Petch's constants for some metals and alloys

|                    | structure       | $\sigma_0$ [MPa] | $k$ [Mpa · m <sup>1/2</sup> ] |
|--------------------|-----------------|------------------|-------------------------------|
| Cu                 | FCC             | 25               | 0.11                          |
| Ti                 | EC              | 80               | 0.40                          |
| mild steel         | BCC             | 70               | 0.74                          |
| Ni <sub>3</sub> Al | L1 <sub>2</sub> | 300              | 1.70                          |

$\sigma_0$ : strength of steels in absence of dislocations

Ni<sub>3</sub>Al is an intermetallic compound used for high temperature applications. It has a very high  $\sigma_0$  because these compounds look like ceramics with very low plastic range but very high  $\sigma_y$ .

## Strengthening of eutectic by refinement in castings



Castings → components obtained after melting and solidification of metals in a mold of specific shape (complex shapes). The resulting component is hard but brittle → it needs strengthening by refinement → a small amount of Strontium is added → modification of the phase diagram.

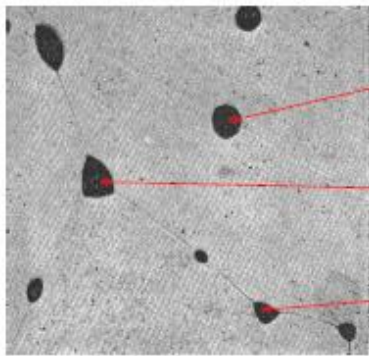
Al-Si alloy → Si is the hard phase.

The structure after solidification can suggest brittleness → crack easily happens because Si (second phase) has a needle-like structure → sharp discontinuity of matrix (the spherical shape is more preferred). Cracks, regards of geometry or distribution may form at the interphase between hard and soft phase. Crack is more severe in needle-like structure.

Higher cooling rate → refinement. It reduces diffusion (grains are fine → low distance).

↑ In the TTT diagram, the alloy is cooled at T lower than before → far from equilibrium, leading to finer microstructure.

#### 4) Strengthening by precipitation



Precipitates within grains

Precipitates at triple points

Precipitates at GBs

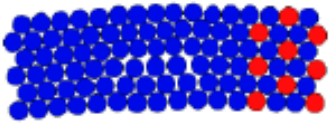
Contributes to strengthening by refinement.

GB precipitation act as a purifying method of the core grain making it more ductile.

Precipitates may be formed into crystals by controlling the conditions of bonding between host metal and guest metal.

Random distribution of atoms → solid solution.

#### Hardening by precipitates

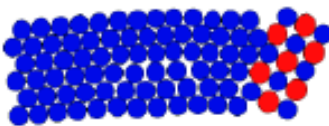
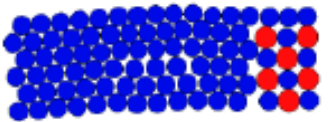


By heat treatment, we can produce precipitates in the solid material.

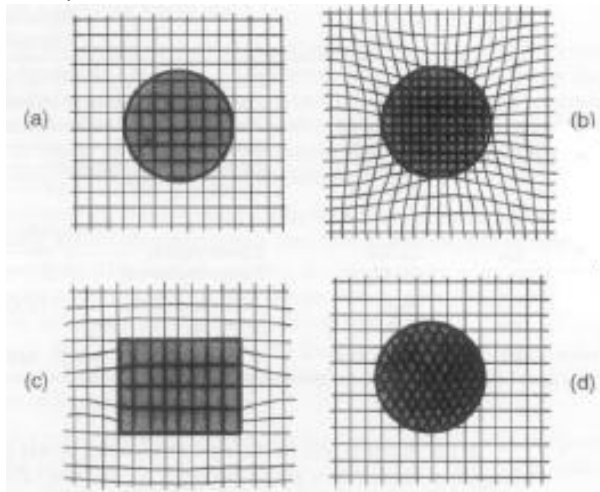
These can have very different composition and crystal structure to the basic metal.

They are very effective blocks to dislocation motion, especially if:

- They are “ordered intermetallic” compounds – directional bonding preferences (e.g.  $\text{CuAl}_2$  in Al alloys,  $\text{Ni}_3\text{Al}$  in Ni-based superalloys,  $\text{Fe}_7\text{Mo}_6$  in maraging steels)
- They are [near] covalent compounds: e.g.  $\text{Fe}_3\text{C}$ , WC,  $\text{AlN}$  in steels,  $\text{MoSi}_2$  in Al alloys
- They have a distinct interface with the “matrix” metal
- They have an associated stress field
- They are large, and thus difficult to cut through (however, larger means fewer and more widely spaced).



#### Precipitate's coherence



**(a) Coherent** → Crystal lattice of precipitate coincides with the crystal lattice of matrix – continuation of lattice planes → Ideal, never happens in reality. Every time a precipitate is formed there is always stress interaction between precipitates and matrix

**(b) Coherent with lattice strain** → in fact we have distortion of the matrix. Distortion increases with decreasing of the size of the precipitate

**(c) Semi coherent: partial lattice continuity** → change of the shape is irrelevant

**(d) Incoherent: lattice discontinuity** → Lattice of precipitates is completely different from lattice of matrix → no stress interaction between matrix and precipitate.

It happens when precipitate is quite large.